



1. Python Basics (Beginner | Programming)

Course Objective:

Gain a solid foundation in Python programming, mastering core syntax, data structures, and control flows to build simple applications confidently.

By course end, you'll create functional scripts and automate everyday tasks, kickstarting your coding journey.

Total Duration: 10 Hours

Detailed Curriculum Table:

Hour/Module	Topic Name	Key Learning Points
Hour 1	Python Setup & Basics	- Installing Python and IDEs (VS Code, PyCharm) - Variables, data types (int, float, str, bool) - Basic input/output with print() and input()
Hour 2	Control Structures	- Conditional statements (if, elif, else) - Loops (for, while) with break/continue - Simple error handling with try-except
Hour 3	Data Structures I	- Lists: creation, indexing, slicing, methods (append, remove) - Tuples and basic operations - List comprehensions for efficient coding
Hour 4	Data Structures II	- Dictionaries: key-value pairs, methods (get, keys, values) - Sets: unique elements, operations (union, intersection) - Nested structures for real-world data
Hour 5	Functions & Modules	- Defining functions with parameters and return values - Lambda functions and scope (local/global) - Importing modules (math, random)
Hour 6	File Handling	- Reading/writing text files (open, read, write) - CSV handling with csv module - Context managers (with statement) for safe file ops
Hour 7	Strings & Formatting	- String methods (split, join, strip, format) - F-strings and regex basics - Manipulating user inputs effectively
Hour 8	OOP Basics	- Classes, objects, init method - Attributes and methods - Simple inheritance intro
Hour 9	Libraries & Mini-Projects	- Intro to NumPy/Pandas basics - Building a simple calculator or to-do list app

Hour/Module	Topic Name	Key Learning Points
Hour 10	Review & Best Practices	- Code debugging, PEP8 style guide - Virtual environments and package management (pip)

The Final Quiz:

A 30-question multiple-choice and coding challenge quiz (via online platform), covering all modules. Passing criteria: 70% score. Retakes allowed once after review.

Capstone Project:

- **Project Title:** Personal Expense Tracker
- **Problem Statement:** Build a command-line app to track daily expenses, categorize them, and generate simple reports from a CSV file.
- **Required Tech Stack:** Python 3.x, built-in libraries (csv, datetime), basic OOP.
- **Deliverables:** Source code (GitHub repo), demo video (2-3 mins), final report with screenshots.

Certification & Career Path:

Earn **Certificate 1 (Completion)** by finishing all modules and passing the final quiz. Submit your Capstone Project for **Certificate 2 (Project)** upon successful evaluation. Exceptional projects earn an Internship offer at "Build Your Projects," opening doors to junior developer roles and real-world Python applications.

2. Advanced Python (Intermediate | Programming)

Course Objective:

Elevate your Python skills to handle complex applications, data processing, and performance optimization for production-ready code.

You'll build scalable scripts, leverage advanced libraries, and apply best practices for professional software development.

Total Duration: 11 Hours

Detailed Curriculum Table:

Hour/Module	Topic Name	Key Learning Points
Hour 1	Advanced Functions	- Decorators and closures - Generators and iterators (yield) - Recursion with memoization
Hour 2	OOP Deep Dive	- Inheritance, polymorphism, encapsulation - Magic methods (str, len) - Abstract base classes (abc module)
Hour 3	Error Handling & Logging	- Custom exceptions - Logging module setup - Unit testing with unittest/pytest

Hour/Module	Topic Name	Key Learning Points
Hour 4	Data Structures Advanced	- Collections (deque, Counter, defaultdict) - Advanced list/dict comprehensions - Performance analysis (timeit)
Hour 5	Multithreading & Async	- Threading module basics - Asyncio intro (async/await) - Concurrent futures for parallelism
Hour 6	File I/O & Databases	- JSON/XML handling - SQLite integration (sqlite3) - Context managers for resources
Hour 7	NumPy & Pandas	- NumPy arrays, vectorization - Pandas DataFrames (filtering, grouping) - Data cleaning and visualization (matplotlib)
Hour 8	Web Scraping & APIs	- Requests and BeautifulSoup - API calls (REST, JSON parsing) - Rate limiting and headers
Hour 9	GUI Development	- Tkinter basics (windows, widgets) - Event-driven programming - Simple app deployment
Hour 10	Optimization & Deployment	- Profiling (cProfile) - Virtualenvs, requirements.txt - Docker basics for Python apps
Hour 11	Capstone Prep & Review	- Code reviews, design patterns - Building portfolio projects

The Final Quiz:

40-question mix of MCQs, code snippets, and debugging tasks. Passing criteria: 75% score. Includes live coding segment; one retake available.

Capstone Project:

- **Project Title:** Automated Stock Analyzer Dashboard
- **Problem Statement:** Develop a GUI app that fetches stock data via API, analyzes trends with Pandas/NumPy, and visualizes insights.
- **Required Tech Stack:** Python 3.x, Pandas, NumPy, Matplotlib, Tkinter, Requests.
- **Deliverables:** Executable app (GitHub), usage guide, performance report.

Certification & Career Path:

Secure **Certificate 1 (Completion)** via all modules and quiz success. **Certificate 2 (Project)** follows Capstone approval. Exceptional submissions qualify for an Internship at "Build Your Projects," accelerating paths to backend developer or data analyst positions.

3. Prompt Engineering (Beginner | AI)

Course Objective:

Master crafting effective prompts for AI models like GPT to generate accurate, creative outputs for real-world tasks.

Unlock AI's potential to boost productivity in writing, coding, and problem-solving without needing deep technical expertise.

Total Duration: 8 Hours

Detailed Curriculum Table:

Hour/Module	Topic Name	Key Learning Points
Hour 1	AI Basics & Prompting Intro	- What is prompt engineering? - LLM fundamentals (tokens, context) - Playground tools (ChatGPT, Grok)
Hour 2	Core Techniques	- Zero-shot, few-shot, chain-of-thought prompting - Role-playing and personas - Specificity and clarity rules
Hour 3	Structuring Prompts	- Prompt templates and chaining - Delimiters, examples in prompts - Handling ambiguity
Hour 4	Advanced Patterns	- Tree-of-thoughts, self-critique - Output formatting (JSON, tables) - Iterative refinement
Hour 5	Domain-Specific Prompts	- Coding assistance prompts - Creative writing and ideation - Data analysis prompts
Hour 6	Tools & APIs	- OpenAI API basics (Python integration) - Prompt chaining in code - Cost optimization
Hour 7	Evaluation & Best Practices	- Prompt testing frameworks - Bias mitigation, ethical considerations - A/B testing prompts
Hour 8	Hands-On Projects	- Building prompt libraries - Real-world case studies

The Final Quiz:

25-question practical quiz with prompt-writing challenges and evaluation. Passing criteria: 70% score. Feedback provided instantly.

Capstone Project:

- **Project Title:** AI Content Generator Toolkit
- **Problem Statement:** Create a prompt library and simple script for generating blog posts, code snippets, or marketing copy on demand.
- **Required Tech Stack:** OpenAI API (or free alternatives), Python basics.

- **Deliverables:** Prompt templates (GitHub), demo script, evaluation metrics report.

Certification & Career Path:

Certificate 1 (Completion) for module completion and quiz pass. Certificate 2 (Project) upon Capstone review. Top "Exceptional" projects win an Internship at "Build Your Projects," ideal for AI prompt specialist or content roles.

4. Introduction to AI & ML (Beginner | AI | Machine Learning)

Course Objective:

Demystify AI and ML fundamentals, from data prep to model deployment, enabling you to build and deploy simple intelligent systems.

Transform ideas into working ML prototypes using accessible tools, paving the way for data-driven projects.

Total Duration: 12 Hours

Detailed Curriculum Table:

Hour/Module	Topic Name	Key Learning Points
Hour 1	AI/ML Overview	- AI vs ML vs DL definitions - Real-world applications - Supervised/unsupervised/reinforcement learning
Hour 2	Python for ML Setup	- Anaconda, Jupyter Notebooks - NumPy/Pandas essentials - Scikit-learn intro
Hour 3	Data Preprocessing	- Loading/cleaning data - Handling missing values, scaling - Train-test splits
Hour 4	Linear Regression	- Model math basics - Training/evaluation (MSE, R2) - Boston housing example
Hour 5	Classification Basics	- Logistic regression, decision trees - Metrics (accuracy, precision, recall) - Iris dataset hands-on
Hour 6	Ensemble Methods	- Random forests, boosting (XGBoost) - Overfitting prevention - Hyperparameter tuning
Hour 7	Clustering & Dimensionality	- K-means, PCA - Unsupervised learning use cases - Elbow method
Hour 8	Neural Networks Intro	- Perceptrons to simple NNs - Activation functions, backpropagation - Keras/TensorFlow basics
Hour 9	Model Evaluation	- Cross-validation, ROC curves - Bias-variance tradeoff - Feature selection

Hour/Module	Topic Name	Key Learning Points
Hour 10	Deployment Basics	- Pickle/Joblib for models - Streamlit for web apps - Cloud intro (Google Colab)
Hour 11	Ethics & Projects	- Fairness, interpretability - Mini ML projects review
Hour 12	Capstone Prep	- End-to-end workflow - Portfolio building

The Final Quiz:

35 questions: theory MCQs + model-building tasks. Passing criteria: 70% score. Includes Jupyter notebook submission.

Capstone Project:

- **Project Title:** Customer Churn Predictor
- **Problem Statement:** Build an ML model to predict customer churn from telecom data, with a simple web dashboard.
- **Required Tech Stack:** Scikit-learn, Pandas, Streamlit.
- **Deliverables:** Trained model (GitHub), Streamlit app, accuracy report.

Certification & Career Path:

Certificate 1 (Completion) after modules and quiz. Certificate 2 (Project) post-Capstone. Exceptional work secures an Internship at "Build Your Projects," launching ML engineer careers.

5. AI + Blockchain (Advanced | Web3 & AI)

Course Objective:

Integrate AI with blockchain for innovative Web3 apps, from smart contracts enhanced by ML to decentralized AI agents.

Develop cutting-edge dApps that leverage AI predictions on-chain, positioning you at the forefront of Web3 innovation.

Total Duration: 9 Hours

Detailed Curriculum Table:

Hour/Module	Topic Name	Key Learning Points
Hour 1	Blockchain & Web3 Basics	- Ethereum, Solidity intro - Smart contracts, EVM - Wallets (MetaMask), Web3.py
Hour 2	AI on Blockchain	- Oracles (Chainlink) for AI data - Decentralized ML concepts - Use cases (NFT generation, DeFi predictions)

Hour/Module	Topic Name	Key Learning Points
Hour 3	Smart Contracts with AI	- Solidity basics (functions, events) - Integrating AI APIs (via oracles) - Gas optimization
Hour 4	Web3 Development	- Hardhat/Truffle setup - Deploying contracts (testnets) - Frontend integration (React, ethers.js)
Hour 5	ML Models for Blockchain	- On-chain ML inference limits - Off-chain AI with on-chain verification - TensorFlow.js for browser ML
Hour 6	Advanced Integrations	- IPFS for decentralized data storage - AI-driven NFTs (generative art) - Zero-knowledge proofs with AI
Hour 7	DeFi + AI	- Yield prediction models - Automated trading bots on-chain - Risk assessment smart contracts
Hour 8	Security & Scaling	- Common vulnerabilities (reentrancy) - Layer 2 solutions (Polygon) - Auditing AI-blockchain systems
Hour 9	Deployment & Projects	- Full-stack dApp build - Portfolio and launch strategies

The Final Quiz:

30 advanced questions: code reviews, scenario-based. Passing criteria: 80% score. Solidity coding required.

Capstone Project:

- **Project Title:** AI-Powered NFT Marketplace Predictor
- **Problem Statement:** Create a dApp where AI predicts NFT value trends, mints dynamic NFTs, and executes trades via smart contracts.
- **Required Tech Stack:** Solidity, Web3.py/ethers.js, TensorFlow.js, Chainlink, IPFS.
- **Deliverables:** Deployed contract (testnet), frontend dApp (GitHub), demo video, security audit summary.

Certification & Career Path:

Certificate 1 (Completion) for modules and quiz. **Certificate 2 (Project)** after evaluation.

"Exceptional" projects offer an Internship at "Build Your Projects," targeting Web3 AI developer roles in high-demand sectors.